ModelIt!

Student Activity Pack

Can Soil Save the Climate?

Modeling Soil Carbon Sequestration, Regenerative Agriculture, and Feedback Loops

Grade 11L2 | NGSS HS-LS2-4, HS-ESS3-3 | Stability and Change / Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think happens to the carbon in a dead leaf after it falls to the ground — where does it go?**

2. **Why might the way we farm affect how much CO2 is in the atmosphere?**

3. **Draw a diagram showing how carbon might move between the atmosphere, plants, and soil.**

4. **Do you think soil can act like a sponge for carbon pollution? What would make it absorb more or less?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Agricultural Management Practice
- Plant Carbon Fixation Rate
- Soil Microbial Activity
- Soil Organic Carbon Stock
- Soil Respiration Rate
- Net Soil Carbon Balance

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Conventional Baseline

Tillage-based farming, bare fallow, synthetic inputs

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Regenerative Transition

No-till, cover crops, diverse rotations adopted

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Climate Warming Feedback

Regenerative practices plus 2-3°C warming

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Permafrost Carbon Pool — Arctic permafrost contains approximately 1,500 gigatons of frozen organic carbon — nearly twice the atmospheric total — which is beginning to thaw and decompose as global temperatures rise, creating a massive potential positive feedback to climate warming

Soil Biodiversity Index — The diversity and abundance of soil organisms from bacteria and fungi to earthworms and arthropods — higher biodiversity increases the efficiency of carbon stabilization and nutrient cycling, while degraded soils with low biodiversity process carbon less effectively

Water Holding Capacity — Soil organic carbon dramatically improves water retention — each 1% increase in soil organic matter allows soil to hold an additional 75,000 liters of water per hectare, creating a powerful co-benefit linking carbon sequestration to drought resilience and flood prevention

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Could changing how we farm pull enough carbon out of the atmosphere to slow climate change — or is soil carbon sequestration too good to be true? Use evidence from your simulation to support your answers.

1. **Why does your model show that soil carbon sequestration has a 'saturation point' — and what does this mean for its role as a long-term climate solution?**

2. **How does the soil-climate feedback loop in your model reveal a potential tipping point where soils switch from helping to hurting climate stability?**

3. **What does the model reveal about the difference between the speed at which soil carbon can be lost (years through tillage) versus gained (decades through regeneration)?**

4. **How do the co-benefits of soil carbon — water retention, biodiversity, crop resilience — change the cost-benefit calculation of regenerative agriculture?**

5. **What does your model suggest about who should pay for soil carbon sequestration — farmers, consumers, governments, or carbon markets — and why?**

8 STEM Challenge: Design a Regional Soil Carbon Sequestration Program

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YOUR ENGINEERING MISSION

Design a farm-scale regenerative agriculture transition program that maximizes soil carbon sequestration while maintaining crop yields and economic viability for farmers over a 25-year implementation period.

THE SCENARIO

A state agricultural department wants to convert 100,000 hectares of conventional cropland to regenerative practices as part of its climate action plan. Your team has been contracted to design the transition program, including practice recommendations, carbon measurement protocols, farmer incentive structures, and realistic sequestration projections.

Engineering Design Thinking

1. WHAT SOIL SAMPLING AND ANALYSIS PROTOCOLS WOULD YOU USE TO ESTABLISH BASELINE CARBON STOCKS AND TRACK CHANGES OVER TIME?

2. HOW WOULD YOU PHASE THE TRANSITION TO MINIMIZE ECONOMIC DISRUPTION WHILE MAXIMIZING EARLY CARBON GAINS?

3. WHAT COMBINATION OF PRACTICES (NO-TILL, COVER CROPS, COMPOST, ROTATIONAL GRAZING) WOULD YOU RECOMMEND FOR DIFFERENT SOIL TYPES AND FARM OPERATIONS?

4. HOW WOULD YOU ACCOUNT FOR THE PERMANENCE PROBLEM — THE RISK THAT STORED CARBON IS RELEASED IF FARMERS REVERT TO CONVENTIONAL PRACTICES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Soil Scientists study the physical, chemical, and biological properties of soils to understand their role in ecosystems, agriculture, and climate regulation. They work for the USDA, universities, agricultural consulting firms, and environmental organizations, earning $65,000–$120,000/year. Carbon Market Analysts who verify and trade agricultural carbon credits at environmental finance firms earn $80,000–$160,000/year.